Heat Treatment Edu Kit 1.1 – Complete User Manual

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1. Introduction

Heat Treatment Edu Kit 1.1 is an integrated educational software suite designed for materials science and engineering students, researchers, and practising heat treatment specialists. The package combines three powerful modules:

- **Temperdata** – Analysis and visualisation of steel tempering curves (hardness vs. tempering temperature for different tempering times).
- **Jomina Analyzer** – Prediction of steel hardenability using ASTM/Grossman, Just, and de Cremona methods, plus Jominy end-quench curve simulation and experimental data fitting.
- **Quenching Studio** – Modelling of phase transformations during continuous cooling (ferrite, pearlite, bainite, martensite) and generation of TTT/CCT diagrams based on steel composition.

All modules are built with a user-friendly PyQt5 graphical interface and rely on proven empirical models from peer-reviewed literature. The suite is released under the GNU General Public License v3 and is intended for educational and non-commercial use.

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2. System Requirements and Installation

2.1 Minimum Requirements

- Operating System: Windows 10/11 (64-bit), Linux, or macOS.
- Processor: 2 GHz dual-core.
- RAM: 4 GB (8 GB recommended for large datasets).
- Hard Disk: 500 MB free space.
- Python 3.8 – 3.12 if running from source; pre-compiled executables require no Python installation.

2.2 Installing from Source (Python)

1. Install Python 3.11 or 3.12.
2. Install required packages:

   bash

   ```
   pip install numpy scipy matplotlib pandas PyQt5
   ```

3. Download the file `Heat treatment edu kit 1.1.py` and the example CSV file `Tempering data for carbon and low alloy steels - Raiipa.csv`.
4. Run the program:

   bash

   ```
   python "Heat treatment edu kit 1.1.py"
   ```

2.3 Using a Pre-compiled Executable

- Place the `.exe` file and the CSV data file in the same folder.
- Double-click the executable – no additional installation is needed.

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3. Getting Started – Launcher Window

When you start the application, a launcher window appears with three large buttons:

- **Open Temperdata (Tempering Analysis)**
- **Open Jomina Analyser (Hardenability)**
- **Open Quenching Studio (Phase Transformations)**

Click any button to open the corresponding module. You can have multiple modules open simultaneously; each opens in its own window.

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4. Module 1: Temperdata – Steel Tempering Curve Analysis

4.1 Purpose and Theory

Tempering is a heat treatment process applied to quenched steels to reduce brittleness and improve toughness. The relationship between tempering temperature, holding time, and final hardness is essential for selecting proper heat treatment parameters.

Temperdata visualises experimental data points (hardness vs. temperature) for different tempering times. It allows interactive exploration of the curves, displays steel chemical composition, and exports plots and tables.

4.2 User Interface

The module contains four tabs:

- **Plot** – Main graph area with hovering tooltips.

- **Steel Information** – Shows chemical composition and basic statistics.
- **Data Table** – Tabular view of the loaded tempering data.
- **Description & Manual** – Program information and license.

4.3 Loading Data

- Use the **Load CSV Data** button to select a CSV file.
- The CSV must contain at least these columns: `Steel type`, `Tempering temperature (ºC)`, `Final hardness (HRC) - post tempering`, and optionally `Tempering time (s)` and chemical composition columns (`C (%wt)`, `Mn (%wt)`, ...).
- If a file named `Tempering data for carbon and low alloy steels - Raiipa.csv` exists in the same folder, it loads automatically on startup.

4.4 Visualising Tempering Curves

- After loading, select a steel grade from the drop-down list.
- The plot shows hardness vs. temperature; each tempering time appears as a separate curve with markers.
- **Hover** your mouse over any data point – a tooltip displays time, temperature, and hardness.
- Use the built-in matplotlib toolbar (zoom, pan, save as image) at the bottom of the plot.

4.5 Steel Information and Data Table

- The **Steel Information** tab lists chemical elements (C, Mn, Si, Ni, Cr, Mo, V, Al, Cu) if present in the CSV. It also shows the hardness and temperature ranges.
- The **Data Table** tab presents the data sorted by time and temperature. Column headers: Time [s], Temperature [°C], Hardness [HRC], Source (if available).

4.6 Saving Results

- **Save Plot** – Exports the current graph as PNG, JPG, PDF, or SVG.
- **Save Table to CSV** – Exports the currently displayed table (filtered by steel type) to a CSV file.

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5. Module 2: Jomina Analyzer – Hardenability Calculations

5.1 Theoretical Background

Hardenability is the ability of a steel to form martensite during quenching. It is measured by the **ideal critical diameter (DI)**. The Jomina Analyzer implements three empirical DI formulas:

- **ASTM/Grossman** – Uses multiplication factors for alloying elements (Mn, Si, Ni, Cr, Mo, Cu) and a carbon-dependent base DI.
- **Just (1969)** – A simplified model with factors for C, Mn, Cr, Mo, Ni, V.
- **de Cremona (1970)** – Extended version that includes Si, Cu, V, W.

Additionally, the module can:

- Simulate Jominy end-quench hardness profiles using the Just method and the ASTM A255 method.
- Estimate the actual DI from experimental Jominy data (hardness vs. distance).
- Calculate the ideal DI (for a quenching severity factor $H=\infty$).

5.2 Main Interface: DI Calculation Tab

The first tab is divided into three panels:

Left panel – Steel Selection and Composition

- Choose a predefined steel from a database (e.g., 30CrNiMo8, 42CrMo4, C45).
- Alloy composition (C, Mn, Si, Cr, Ni, Mo, Cu, V, W) and grain size can be edited manually.
- Buttons: **Save Composition** (as JSON), **Load Composition** (JSON).

Middle panel – Steel Information & Heat Treatment

- Displays description, typical heat treatment, and hardness data for the selected steel.

Right panel – DI Results

- Shows calculated DI (in mm) for each method.
- After importing Jominy data, you also see **Experimental (actual)** and **Experimental (ideal)** DI.

Click **Calculate DI** to recompute values when composition changes.

5.3 Jominy Curve Tab – Experimental Data Entry

- The second tab allows you to enter experimental hardness values at standard Jominy distances (1.5, 3, 5, … 50 mm).
- Use **Add Row** / **Remove Row** to adjust the table.
- Checkboxes let you enable/disable up to four experimental curves.
- The **Show Just Method** checkbox overlays the simulated Jominy curve from the Just method.
- Press **Calculate DI from Curve** – the program interpolates the distance at which hardness drops to the "hardness after quenching" value, then converts that distance to a DI using a built-in correlation table.
- The result appears in the DI results panel.

5.4 DI Methods Comparison Tab

- A horizontal bar chart compares the ASTM, Just, and de Cremona DI values.
- Vertical dashed lines mark the actual and ideal experimental DI (if available).
- Use **Update Comparison** to refresh after any changes.

5.5 Saving and Loading Data

- **Save Composition** / **Load Composition** – store alloy compositions in JSON format.
- **Save Jominy Data** – exports the first active experimental curve as CSV (distance, hardness).
- **Load Jominy Data** – imports a CSV with distance and hardness columns and fills the table.
- **Save Results** – saves all DI calculations (ASTM, Just, de Cremona, experimental DI) to a CSV file.
- **Save Plot** – exports the Jominy curve plot as an image.

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6. Module 3: Quenching Studio – Phase Transformations

6.1 Scientific Basis

This module predicts the kinetics of diffusional transformations (ferrite, pearlite, bainite) and athermal martensite formation during continuous cooling. The models are based on:

- **Critical temperatures:** Ae1, Ae3 (Andrews), Bs (Li), Ms (Van Bohemen).
- **Transformation start times** using a modified Johnson-Mehl-Avrami-Kolmogorov approach with sigmoidal functions (`S` and `I` integrals).
- **Hardness** of each phase as a function of cooling rate (empirical equations by Li et al., Van Bohemen).

  The calculations account for ASTM grain size and chemical composition.

6.2 Input Parameters Tab

- **Steel Selection** – Choose from the same database as Jomina Analyzer. Selecting a steel automatically fills composition and grain size.
- **Composition** – 10 elements (C, Mn, Si, Ni, Cr, Mo, V, Co, Cu, W) with double spin boxes.
- **Grain Size** – ASTM number (1–12, default 7).
- **Temperature Parameters** – Initial temperature (Tini, typically austenitising temperature, default 900°C) and final temperature (Tfin, room temperature, default 25°C).
- **Cooling Rates** – Two modes:

1. **Single Cooling Rate** – a specific rate in °C/s.
2. **List of cooling rates** – a multi-line text edit (default values from 1000 down to 0.001 °C/s). Used for hardness vs. cooling rate analysis.

- **Calculation Buttons**:
  - **Calculate Single Cooling Rate**
  - **Calculate Hardness vs Cooling Rate**
  - **Calculate TTT Diagram** (isothermal)
  - **Calculate CCT Diagram** (continuous cooling)

6.3 Single Cooling Rate Analysis

- Enter a cooling rate (e.g., 10 °C/s) and click **Calculate Single Cooling Rate**.
- The program computes the phase fractions versus time and the final phase distribution.
- Results appear in two tabs:
  - **Single Cooling Rate** – two plots: (left) phase fractions vs. time, (right) final phase fractions pie chart.
  - **Single Cooling Rate – Table** – a table summarising cooling rate, final ferrite/pearlite/bainite/martensite fractions, and Vickers hardness.
- **Save results to CSV** button exports the table.

6.4 Hardness vs Cooling Rate

- Enter a list of cooling rates in the text box (one per line) and click **Calculate Hardness vs Cooling Rate**.
- The module calculates final phase fractions and hardness for each rate.
- Two tabs appear:
  - **Hardness vs Cooling Rate** – plots: (left) phase fractions vs. cooling rate (log scale), (right) hardness vs. cooling rate.
  - **Hardness vs Cooling Rate – Table** – tabular results for all rates.
- Export the table to CSV with the provided button.

6.5 TTT and CCT Diagrams

- **TTT Diagram** – shows isothermal transformation curves for 1% and 99% transformation of ferrite, pearlite, and bainite. Horizontal lines indicate Ae3, Ae1, Bs, Ms.
- **CCT Diagram** – continuous cooling transformation diagram with start and finish curves. It also superimposes dashed cooling lines for selected cooling rates (every tenth rate from the list).
- Both diagrams are interactive (zoom, pan, save) via the matplotlib toolbar.

6.6 Saving Results

- Each table (single cooling rate, hardness vs. cooling rate) has a dedicated **Save results to CSV** button.
- The **Save composition** button stores the current alloy and grain size as a JSON file; **Load composition** reloads a previously saved composition.
- Diagrams can be saved using the matplotlib toolbar’s save icon.

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7. File Formats and Data Requirements

7.1 CSV for Temperdata

Required columns (exact spelling):

- `Steel type`
- `Tempering temperature (ºC)`
- `Final hardness (HRC) - post tempering`

- Optional: `Tempering time (s)`, `Source`, chemical composition columns like `C (%wt)`, `Mn (%wt)`, etc.

7.2 CSV for Jominy Experimental Data

Two columns, no header required, but typical first row can be used as header. Example:

```text
Distance (mm),Hardness (HRC)
1.5,58
3,57
...
```

7.3 JSON for Composition

Example:

```json
{
    "composition": {"C":0.34, "Mn":0.6, "Si":0.3, "Cr":1.5, "Ni":1.5, "Mo":0.25, "Cu":0.1, "V":0, "W":0},
    "grain_size": 7
}
```

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8. Troubleshooting and Common Issues

8.1 Module does not open or crashes

- Ensure all required Python packages are installed (`numpy`, `scipy`, `matplotlib`, `pandas`, `PyQt5`).
- If using the compiled executable, check that your antivirus has not quarantined it (false positive possible due to package bundling).

8.2 No data shown in Temperdata

- The CSV must contain the exact column names listed above. Open the file in a text editor to verify.

- If the default file is missing, use the **Load CSV Data** button to manually select a valid file.

8.3 Jominy curve shows no experimental points

- Make sure the distance values in the table are numeric and increasing. Add at least two rows.
- After entering data, click **Calculate DI from Curve** – the plot updates automatically only after that or after toggling checkboxes.

8.4 Quenching Studio – "An error occurred" during calculation

- Check that the cooling rate is positive and not zero.
- Very low cooling rates (<0.001 °C/s) may cause numerical issues; increase the rate slightly.
- The initial temperature (Tini) must be above Ae3; if it is lower, no austenite is assumed, and transformation models will fail.

8.5 Generated plots are empty or missing labels

- This can happen if the composition leads to impossible phase fractions (e.g., C > 0.8%). Adjust composition values to realistic ranges.
- Restart the module if the matplotlib backend becomes corrupted.

8.6 Performance is slow when calculating many cooling rates

- Reduce the number of rates in the list (the default 13 rates is fine). For hundreds of rates, calculations may take several seconds.

8.7 CSV data file not found after compilation with PyInstaller/Nuitka

- If you used `--add-data` or `--include-data-files`, the file is embedded in the executable's temporary folder. The program uses `os.path.exists(...)`, so the file must be present in the working directory. Either copy the CSV next to the `.exe` or modify the code to use `sys._MEIPASS` (PyInstaller) or `__file__` to locate the data.

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9. License and Credits

The Quenching Studio module incorporates transformation kinetics code originally developed by Arthur S. N. (https://github.com/arthursn/transformation-diagrams) and adapted under the GPL v3.

All empirical models are derived from published scientific literature; references can be found in the source code comments.

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10. Appendix – References and Formula Compendium

10.1 Tempering

No built-in formula; purely graphical analysis of user-supplied data.

10.2 Hardenability

- ASTM A255 – Standard Test Methods for Determining Hardenability of Steel.
- Just, E. (1969). "New formulae for calculating hardenability", *Metallurgia*.
- de Cremona, A. (1970). "Factors influencing hardenability", *Metal Progress*.

10.3 Phase Transformations

- Andrews, K.W. (1965). “Empirical formulae for the calculation of some transformation temperatures”, *JISI*.
- Li, M.V. et al. (1998). “A model for the bainite start temperature”, *Metall. Mater. Trans. A*.
- Van Bohemen, S.M.C. (2012). “Modelling of phase transformations in steels”, *Materials Science and Technology*.
- The sigmoidal functions `S` and `I` represent the integral of the transformation rate based on the Avrami equation.

10.4 Hardness prediction

- Maynier, P. et al. (1978). “Hardness prediction from CCT diagrams”, *Hardenability Concepts with Applications to Steel*.

End of Manual

This manual covers all features of Heat Treatment Edu Kit 1.1. For further assistance, please contact the author at m_goral@interia.pl.